Natalyx MVP Fixes

Work Package 10: Intelligence, memory, proposals, and Ask safety

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1 Work Package 10: Intelligence, memory, proposals, and Ask safety

Review draft only. Ownership and sequencing remain recommendations until Luke and Alan approve the plan together.

Plan	Value
Work package	WP10
Plan file	`WP10 - Intelligence, memory, proposals, and Ask safety.md`
Proposed PRs	P10-01, P10-02, P10-03, P10-04
Inspection snapshot	2026-07-12 19:30 America/New_York

1.1 Purpose and outcome

Problem. Private memory, shared case truth, Learn context, and AI proposals have partial contracts but current app code can place IP guide notes into shared memory and provider projections lack provenance enforcement.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned scoped context envelopes, separate private/shared memory revisions, evidence-bound proposal approval, grounded Ask behavior, and cross-service safety evaluation/observability.

Explicit non-goals.

- No LLM writes to case truth
- No medical/legal advice
- No silent fallback success
- No final Learn corpus authoring

1.2 Verified current-state evidence

Classification is explicit: **verified** is observed code/PR/live behavior; **verified limitation** records an evidence gap; **design authority** is an approved planning source; **design decision** is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-014	verified	app / PR #124@84a5a84	app/storage/node0.py:42-200,509-600; tests/python/test_node0.py:441-490	IP/co-IP own candidate reads; GC onboarding writes separate shared/private facts and does not read the IP assertion. The GC cannot see or correct the IP candidate record.
E-015	verified	app / PR #122@e853821	app/storage/facts.py:19-69,107-127	Facts have broad provenance and visibility values but no subject-versus-author role, verification state, derived-from, or supersedes link.
E-016	verified	app / PR #122@e853821	app/storage/proposals.py:117-142	Generic proposal approval permits any case member and can pass caller-selected provenance to create_fact; a dedicated evidence/role rail is required before reviewed/provider-confirmed truth.
E-017	verified	app / PR #124@84a5a84	app/routes/node0.py:668-719	IP Conversation Guide notes are saved as shared memory and sent to intelligence, creating a future cross-role leakage risk.
E-019	verified protected work	app / dirty feature/provider-exchange-layer-v0@84a5a84	app/provider_handoffs.py:109-133 and uncommitted handoff modules	The protected provider work is not a PR. Its broad GC-owner fact selection can include private/provisional data, so only selective port after provenance/purpose review is safe.
E-020	verified	intelligence / origin/main@56f6c4b	src/natalyx_intelligence/case_context.py:8-147; docs/agents/case-context.md	The service returns advisory-only proposals and says durable writes are app-owned; this is a useful boundary but the surrounding context provenance remains incomplete.
E-021	verified	intelligence / origin/main@56f6c4b	src/natalyx_intelligence/ask_natalyx.py:34-42,365-378	Ask accepts known_intake_state without an authorship-class contract and serializes it into model context.

ID	Class	Repository / ref	File, construct or location	Finding
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review - findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.

1.3 Architecture and contracts

- The app owns case truth, authorization, PHI policy, and durable approval; intelligence returns advisory proposals only.
- Every request declares general Learn, private-user, approved-shared-case, or provider-purpose scope; contexts are not merged.
- Failures degrade explicitly with source/boundary notices and no fabricated case answer.

Contract area	Package rule
Ownership boundary	`app`, `intelligence`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Versioned context/advisory contract with strict validation and safe compatibility window. Responses carry source, scope, policy, safety and degraded-mode metadata.
Storage	Revision/supersedes/provenance/evidence fields and the private-note remediation under versioned migrations.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

1.3.1 Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

1.4 Explicit PR sequence

Order	ID	Title	Repository	Dependencies	Gate	Local preparation	Merge readiness
1	P10-01	Version the scoped intelligence context envelope	intelligence	P01-01, P04-02	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P01-01, P04-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval

Order	ID	Title	Repository	Dependencies	Gate	Local preparation	Merge readiness
2	P10-02	Enforce private/shared memory and evidence-bound proposal approval	app	P04-02, P10-01	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P04-02, P10-01; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
3	P10-03	Harden Ask and Learn grounding and degraded-mode behavior	intelligence	P10-01, P11-02	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P10-01, P11-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
4	P10-04	Add intelligence safety evaluations and cross-service observability	intelligence	P10-02, P10-03	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P10-02, P10-03; Gate A0 active and verified; Luke review/final founder approval

1.4.1 P10-01 - Version the scoped intelligence context envelope

Field	Plan
Repository / base	`intelligence` (`main`)
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01, P04-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Dependencies	P01-01, P04-02
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current case_context advisory stub

1.4.1.1 Objective Define and validate versioned request/response envelopes that keep general Learn, private-user memory, approved shared-case facts, and provider-purpose context distinct.

1.4.1.2 Scope

- Explicit scope and audience
- Source/provenance/verification metadata
- Policy/model/version and redacted trace metadata

1.4.1.3 Explicit non-goals

- No durable app write
- No merged contexts

1.4.1.4 Expected files or modules

- `src/natalyx_intelligence/contracts/context.py`
- `src/natalyx_intelligence/case_context.py`
- `docs/agents/case-context.md`
- `tests/test_context_contract.py`

1.4.1.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Versioned context/advisory contract with strict validation and safe compatibility window.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.

Area	Planned change
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.1.6 Tests and validation Objective tests:

- Cross-scope rejection
- unknown version
- private/shared fixtures
- redaction
- service auth

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.1.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.1.8 Rollout Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

1.4.1.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.1.10 Dependencies and parallel work Hard dependencies: P01-01, P04-02. Parallel opportunities: P10-04

1.4.1.11 Objective acceptance criteria

- ☐ Define and validate versioned request/response envelopes that keep general Learn, private-user memory, approved shared-case facts, and provider-purpose context distinct.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.

☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.1.12 Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	FastAPI/Pydantic, AI safety, privacy
Higher-capability review	Mandatory before merge

1.4.2 P10-02 - Enforce private/shared memory and evidence-bound proposal approval

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P10-01; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Dependencies	P04-02, P10-01
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current facts/memory/proposals scaffolds, GC authorship audit

1.4.2.1 Objective Store attributable private/shared memory revisions and require authorized evidence review before an intelligence proposal can affect case truth or provider material.

1.4.2.2 Scope

- Private and shared memory ledgers
- Affected-row inventory for source=node0_conversation_guide_note
- Immediate shared/intelligence exclusion plus idempotent person-owner quarantine/private remediation when rows exist
- Proposal preview/approve/reject
- Evidence/source and reviewer authority
- Audit/outbox

1.4.2.3 Explicit non-goals

- No automatic AI approval
- No guide notes default shared

1.4.2.4 Expected files or modules

- `app/case_intelligence.py`
- `app/storage/memory.py`
- `app/storage/proposals.py`
- `app/storage/facts.py`
- `tests/python/test_case_intelligence.py`

1.4.2.5 Change map

Area	Planned change
Schema/model	Revision/supersedes/provenance/evidence fields and the private-note remediation under versioned migrations.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Role and purpose authorization on every projection/write; ambiguous legacy ownership quarantines rather than broadens access.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.2.6 Tests and validation

Objective tests:

- Zero-row inventory evidence
- affected rows disappear from shared/Ask reads before remediation
- idempotent owner-safe migration
- ambiguous-owner quarantine
- no note content in logs
- private-to-shared denial
- wrong-role approval
- AI provenance remains proposal
- concurrent review
- provider projection

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.2.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.2.8 Rollout Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.2.9 Rollback or forward repair Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.2.10 Dependencies and parallel work Hard dependencies: P04-02, P10-01. Parallel opportunities: P10-03

1.4.2.11 Objective acceptance criteria

- ☐ Store attributable private/shared memory revisions and require authorized evidence review before an intelligence proposal can affect case truth or provider material.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.2.12 Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	AI safety, authorization, data provenance
Higher-capability review	Mandatory before merge

1.4.3 P10-03 - Harden Ask and Learn grounding and degraded-mode behavior

Field	Plan
Repository / base	`intelligence` (`main`)
Gate / size	A2 / Large

Field	Plan
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-01, P11-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Dependencies	P10-01, P11-02
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current ask_natalyx.py and PR #124 Ask integration

1.4.3.1 Objective Answer from the declared scope with approved sources/citations and explicit boundary/failure notices; never fabricate case context or silently fall across scope.

1.4.3.2 Scope

- Approved resource resolution
- Scope-specific prompt/context
- Timeout/unavailable/empty-source behavior
- Citation validation

1.4.3.3 Explicit non-goals

- No medical/legal advice
- No unapproved corpus fallback

1.4.3.4 Expected files or modules

- `src/natalyx_intelligence/ask_natalyx.py`
- `src/natalyx_intelligence/learn_resources.py`
- `tests/test_ask_natalyx.py`
- `docs/agents/ask-natalyx.md`

1.4.3.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Responses carry source, scope, policy, safety and degraded-mode metadata.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.

Area	Planned change
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.3.6 Tests and validation Objective tests:

- Empty/unapproved corpus
- service/provider timeout
- case question without case scope
- citation mismatch
- safety escalation

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.3.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.3.8 Rollout Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

1.4.3.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.3.10 Dependencies and parallel work Hard dependencies: P10-01, P11-02. Parallel opportunities: P10-02

1.4.3.11 Objective acceptance criteria

- ☐ Answer from the declared scope with approved sources/citations and explicit boundary/failure notices; never fabricate case context or silently fall across scope.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.3.12 Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	LLM orchestration, retrieval/citations, safety evaluation
Higher-capability review	Mandatory before merge

1.4.4 P10-04 - Add intelligence safety evaluations and cross-service observability

Field	Plan
Repository / base	`intelligence` (`main`)
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-02, P10-03; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P10-02, P10-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current intelligence auth/telemetry tests

1.4.4.1 Objective Add deterministic safety/grounding evaluations, correlation IDs, latency/failure metrics, and redacted traces across app-intelligence calls.

1.4.4.2 Scope

- Leakage/hallucination/fallback evals
- Correlation and contract version
- Safe logs and alarms

1.4.4.3 Explicit non-goals

- No prompt or participant payload in logs
- No PHI readiness claim

1.4.4.4 Expected files or modules

- `tests/evals/`

- `src/natalyx_intelligence/telemetry.py`
- `docs/runbooks/intelligence.md`
- `app/intelligence_auth.py (consumer follow-up PR if needed)`

1.4.4.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.4.6 Tests and validation Objective tests:

- PII/PHI redaction fixtures
- correlation continuity
- failure classification
- model fallback
- eval threshold

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.4.7 Diagnostics and observability Metrics and redacted structured events; app-side consumer change must remain a separate coordinated PR if repository boundaries require it.

1.4.4.8 Rollout Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

1.4.4.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.4.10 Dependencies and parallel work Hard dependencies: P10-02, P10-03. Parallel opportunities: P10-01

1.4.4.11 Objective acceptance criteria

- ☐ Add deterministic safety/grounding evaluations, correlation IDs, latency/failure metrics, and redacted traces across app-intelligence calls.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.4.12 Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	AI evaluation, observability, privacy
Higher-capability review	Mandatory before merge

1.5 Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`intelligence`	advisory inference, scoped response/citation contract; no case truth

Merge order inside this package: `P10-01` → `P10-02` → `P10-03` → `P10-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

1.6 Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Scope isolation; Proposal approval/rejection; Citation/source validity; Unavailable service; Prompt/context leakage; Trace redaction	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

1.7 Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-AI-01	Founder scope/cost final; Alan later; PHI externally gated	Luke sets scope and cost; Alan decides implementation later	Alan - later before P10-01	P10-01, P10-02, P10-03, P12-01; Before P10-01	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.

1.7.1 D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.

Field	Recorded decision or requirement
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

1.7.2 D-AI-01 - Intelligence context and approval authority matrix

Field	Recorded decision or requirement
Decision state	Final founder policy; PHI externally dependent
Founder outcome	AI is bounded and advisory: role-scoped education, retrieval, summaries, question drafts, next steps and provider preparation using approved Learn and authorized source-labelled context. It never diagnoses, ranks, approves, clears or writes durable truth; authorized human approval precedes memory, case or provider material.
Founder constraints	Deterministic or curated fallback; $10 per active case/month and $100 total beta/month ceilings with alerts and graceful fallback; No PHI before B2
What remains for Alan	Choose Bedrock model, prompts, retrieval, schemas, fallback, human-review rails and cost controls
Evidence required	Scope/authority negative tests; Fallback and malformed-response tests; Safety and cost evidence
Approval rule	Luke approves broader AI authority, higher cost or any PHI use; B2 external approval is mandatory for PHI.
External dependencies	B2 external security/privacy/compliance approval before PHI
Affected PRs / latest point	P10-01, P10-02, P10-03, P12-01; Before P10-01
Programme effect	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
Independent recommendation	Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.

Implementation risk	Owner	Treatment
Private-to-shared leakage	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Hallucinated truth	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Sensitive logs	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Contract version drift	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

1.8 Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P10-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P10-02	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P10-02	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
main (behind origin/main)	9d824e8	dirty Learn registry/corpus plus uv.lock	P11-02/P10-03 as appropriate	Selective port from protected worktree	Do not pull/reset/stash; extract in isolated worktree later.

Package source inventory:

- Intelligence case_context advisory stub
- App proposal/fact/memory scaffolds
- Protected intelligence Learn corpus changes

1.9 Ownership recommendation

Alan implements service/app contracts; Luke approves truth/privacy policy. Higher-capability, security/privacy, and clinical/legal safety review required.

PR	WP DRI	Primary implementer	Technical-decision owner	Reviewer	Final approver	Exception
P10-01	Alan	Alan (proposed)	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism	Luke	Luke	None
P10-02	Alan	Alan (proposed)	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism	Luke	Luke	None
P10-03	Alan	Alan (proposed)	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism	Luke	Luke	None
P10-04	Alan	Alan (proposed)	None; implement the settled founder constraints and approved contracts	Luke	Luke	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

1.10 Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and review record complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- ☐ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- ☐ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- ☐ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- ☐ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- ☐ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.